Dario Loi

Boston, MA | dario-loi.github.io | github.com/dario-loi | Google Scholar

Research Interests: Topological deep learning, sheaf neural networks, graph neural networks, simplicial complexes

EDUCATION

Northeastern University September 2026 – Present
Ph.D. in Computer Engineering, MENTIS Lab, advised by Prof. Francesco Restuccia *Boston, MA*

Sapienza University of Rome September 2023 – October 2025
M.S. in Computer Science *Rome, Italy*
GPA 29.8/30 (3.98/4.0), 110/110 cum laude; Honors Program with Prof. Iacopo Masi

Sapienza University of Rome September 2020 – July 2023
B.S. in Applied Computer Science and Artificial Intelligence *Rome, Italy*
GPA 28.4/30 (3.79/4.0), 110/110 cum laude; Honors Program with Prof. Danilo Avola

PUBLICATIONS

Implicit Inversion turns CLIP into a Decoder
A. D'Orazio, M.R. Briglia, D. Crisostomi, **D. Loi**, E. Rodolà, I. Masi
International Conference on Learning Representations (ICLR), 2026

- Ran the hyperparameter searches and produced qualitative results

A Metacognitive Framework for Test-Time Adaptation: Taxonomy and Research Challenges
D. Loi, G. Onorato, A. Coviello, S. Rifat, N.D. Bastian, F. Restuccia
AAAI Fall Symposium on Artificial Metacognition (METACOG-26), 2026

- Built the TTA taxonomy; ran the experiments in which deliberate cues admitted up to 22.1x fewer OOD samples

Implicit Sheaf Operators for Effective and Efficient Cellular Sheaf Learning
G. Onorato, **D. Loi**, C. Apostol, N.D. Bastian, F. Restuccia
Preprint, 2026; under review at AAAI-27

- Designed Sheaf Convolutional Networks (up to +3.93% accuracy over NSD) and proved complexity bounds

EXPERIENCE

Research Intern December 2025 – June 2026
Evidence Srl *Pisa, Italy*

- Prototyped an agentic LLM for Simulink-style power plant control; scaled reliable output from 5 to 100+ blocks

Teaching Assistant May 2024 – September 2024
Computer Architecture and Operating Systems, Sapienza University of Rome *Rome, Italy*

Computer Vision Software Engineer October 2023 – June 2024
Sapienza Flight Team *Rome, Italy*

- Built real-time YOLOv8 detection of payload drop sites from a drone in flight (SUAS 2024)

OPEN SOURCE AND PROJECTS

Topological Deep Learning Challenge 2026 (TopoBench) github.com/geometric-intelligence/topobench

- Implemented the winning GNN-track entry, Directed Sheaf NNs (Fiorini et al., 2026), and three more sheaf models
- Co-authored the peer-reviewed challenge reports at TAG-DS 2025 and 2026 (PMLR)

Graph Neural Networks for Next-POI Recommendation (M.S. Project) github.com/dario-loi/foursquare

- Designed a transformer-based GNN in PyTorch Lightning: +2.3% top-1, +13.7% top-20 accuracy over prior methods

HONORS AND AWARDS

- 1st of 37 in Track 1 and Excellent Portfolio honorable mention, Topological Deep Learning Challenge 2026
- Top 3 of 27 papers, AAAI Fall Symposium on Artificial Metacognition (METACOG-26) 2026
- Honors Program, Sapienza University of Rome: B.S. (one of eight selected) and M.S. 2022 – 2025

SKILLS

Programming: Python, C++, C, CUDA, Java, SQL; Git, Linux, Docker, SLURM, Weights & Biases
Machine Learning: PyTorch, PyTorch Geometric, TopoX, Hugging Face, LangGraph, LangChain, vLLM